

ENGINEERING FLOW GRAPHICS

=====+
| MEDBRIDGE+ SMART HEALTH ID |
+=====+
=====+

Unique AI Sequence

Birth Year

Birth Day & Month

Birth Place Code

```

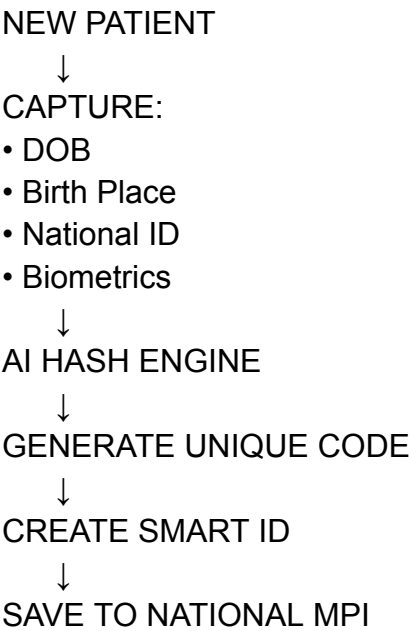
=====
=====+
| PATIENT DATA                               |
+=====
=====+

```

FINAL MEDBRIDGE ID:
KG-0508-24-721

3. ID CREATION ENGINE

```
+=====+
=====+
|          MEDBRIDGE+ ID GENERATION ENGINE          |
+=====+
=====+
```



4. DATABASE STRUCTURE

```
+=====+
=====+
| TABLE: PATIENTS                                |
+=====+
=====+
```

medbridge_id	KG-0508-24-721	
national_id	1199880076543210	

first_name	Uwase	
last_name	Aline	
birth_place	Kigali	
date_of_birth	2024-08-05	
biometric_hash	encrypted_hash	

5. NATIONAL DISTRICT PREFIXES

+=====+
| RWANDA BIRTH PLACE PREFIXES |
+=====+

- KG → Kigali
- MS → Musanze
- HY → Huye
- RB → Rubavu
- BG → Bugesera
- RW → Rwamagana
- KY → Kayonza
- NY → Nyagatare
- RZ → Rusizi
- MH → Muhanga

6. HASH GENERATION LOGIC

+=====+
=====+
| AI UNIQUENESS ENGINE |
+=====+
=====+

INPUT:

- National ID
- Birth Place
- DOB
- Timestamp
- Facility ID

↓ SHA256 HASH ↓

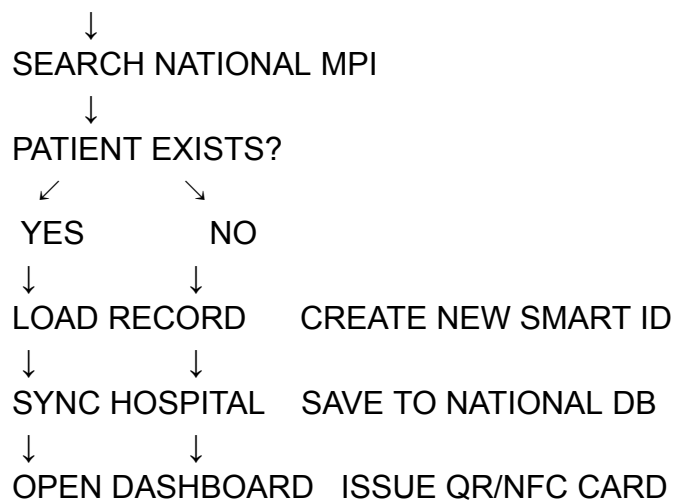
OUTPUT:

721

7. FINAL GENERATION FLOW

```
+=====+
|          COMPLETE NATIONAL GENERATION FLOW          |
+=====+
=====+
```

PATIENT ARRIVES



8. OFFLINE MODE

```
+=====+
=====+
| OFFLINE-FIRST MODE                                     |
+=====+
=====+
```

NO INTERNET



CREATE TEMP ID

TMP-KG-8821



LOCAL STORAGE



INTERNET RETURNS



SYNC TO NATIONAL MPI



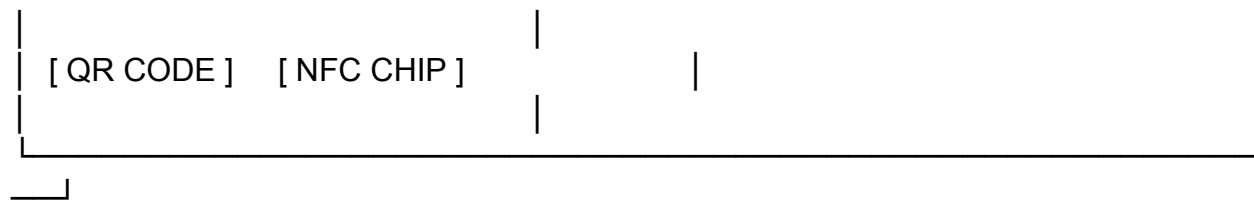
FINAL ID GENERATED

KG-0508-24-721

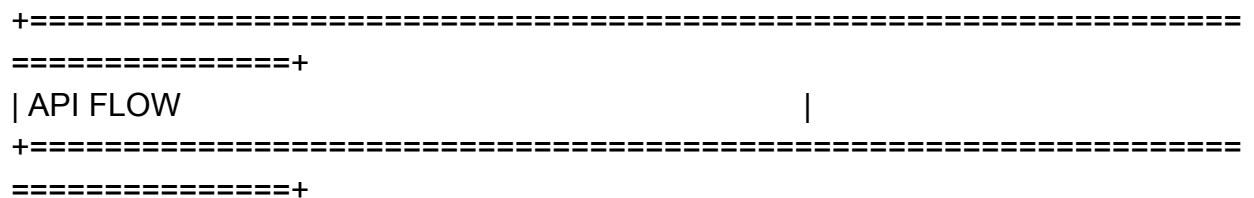
9. QR + NFC HEALTH CARD

```
+=====+
=====+
| MEDBRIDGE+ SMART HEALTH CARD                         |
+=====+
=====+
```

REPUBLIC OF RWANDA	
MEDBRIDGE+ HEALTH CARD	
NAME: Uwase Aline	
ID: KG-0508-24-721	

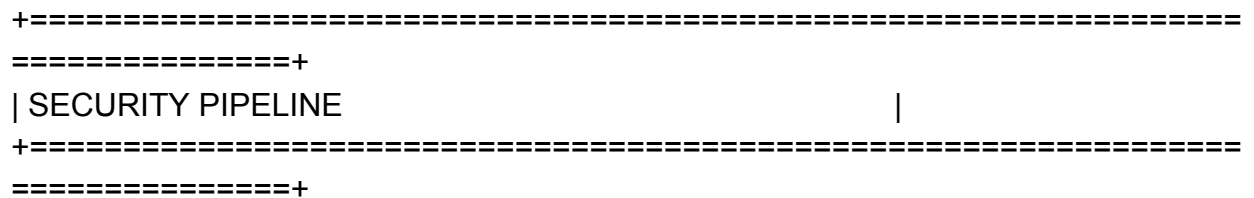


10. ENGINEERING API FLOW

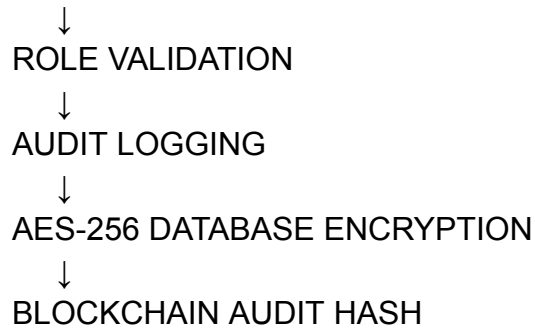


POST /patients/register
↓
VALIDATE DATA
↓
CHECK NATIONAL MPI
↓
GENERATE SMART ID
↓
SAVE TO POSTGRESQL
↓
RETURN RESPONSE

11. SECURITY FLOW



TLS 1.3
↓
JWT AUTH



12. FINAL SYSTEM ARCHITECTURE

